

สถาปัตยกรรมอากาศยานไร้คนขับ และหลักอากาศพลศาสตร์พื้นฐาน

UAV Architecture & Aerodynamics of Drone

19 September 2026 @ Tassana Garden



วิทยากร : คุณท๊อป | T Maker





AGENDA

UAV Architecture

- What is a UAV ?
- Type of UAV
- Quadcopter

Principles of Drone

- Drone Dynamics
- Communication System
- Flight controller
- Flight Control Loop
- PID Control

Assembly & Setup

- SimpleFly Learning Kit
- Firmware Installation
- Setup & Calibration
- Tuning
- Manual Control



What is a UAV ?

A UAV stands for Unmanned Aerial Vehicle. It's essentially an aircraft that flies **without a human pilot on board**. UAVs are controlled either remotely by a person (via radio signals, computers, or controllers) or autonomously using pre-programmed flight plans and onboard sensors.



Source: wikipedia.org/wiki/Unmanned_aerial_vehicle



Type of UAV

Fixed wing



Source: [U.S. Air Force website](#)

Single rotor



Source: [velos-rotors.com](#)

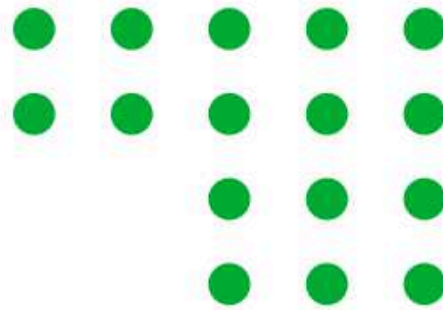
Bicopter



Source: [www.d-botix.com](#)



Type of UAV



Tricopter



Source:
[www.britannica.com/technology/
unmanned-aerial-vehicle](http://www.britannica.com/technology/unmanned-aerial-vehicle)

Quadcopter



Source: www.t-drones.com

Hexacopter



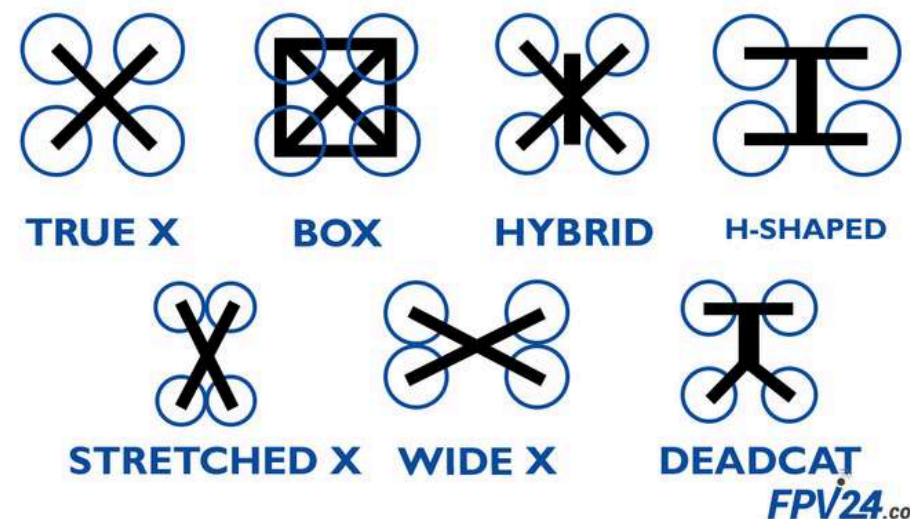
Source: www.viewprouav.com



Quadcopter

The quadcopter became the standard because it balances torque with four fixed motors alone — no complex mechanics. This makes it stable, affordable, and easy to fly, from toys to professional drones

COMMON FPV FRAME SHAPES



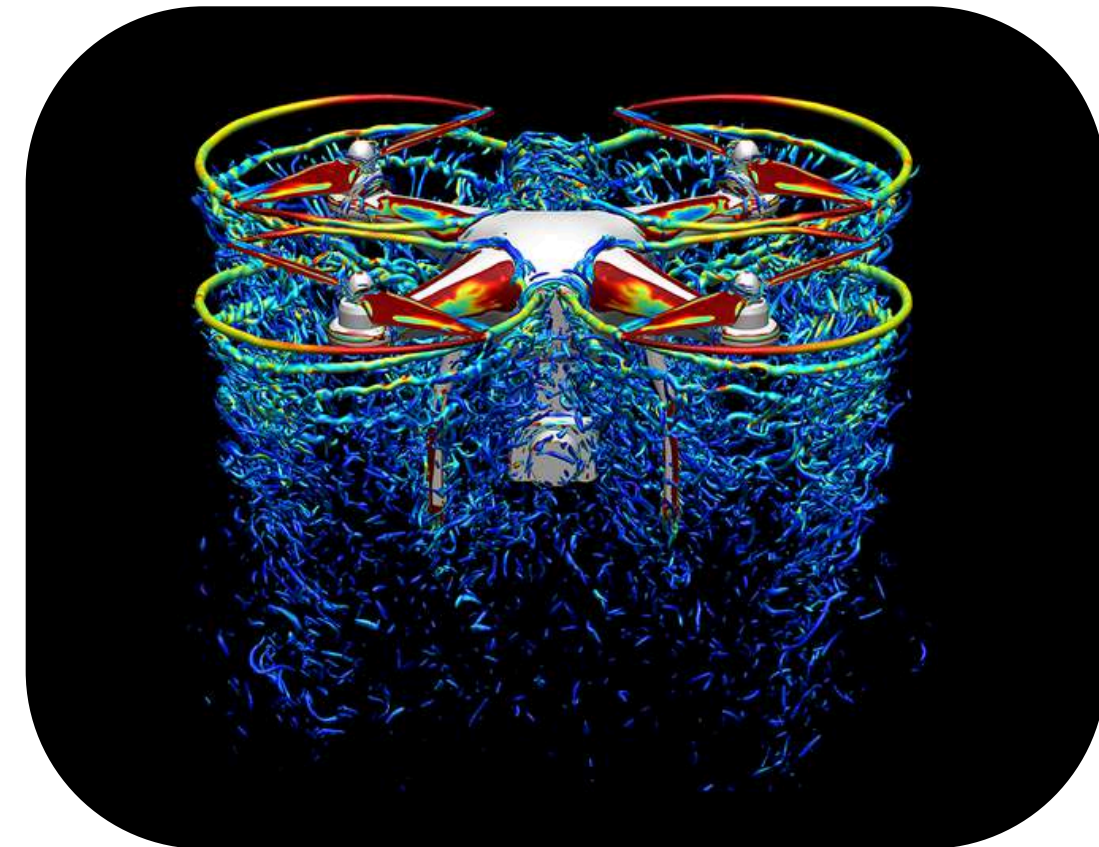
Source: www.fpv24.com





Drone Dynamics

- How Drones Generate Lift
- Forces and Torques
- Frames of Reference
- Degrees of Freedom (DOF)



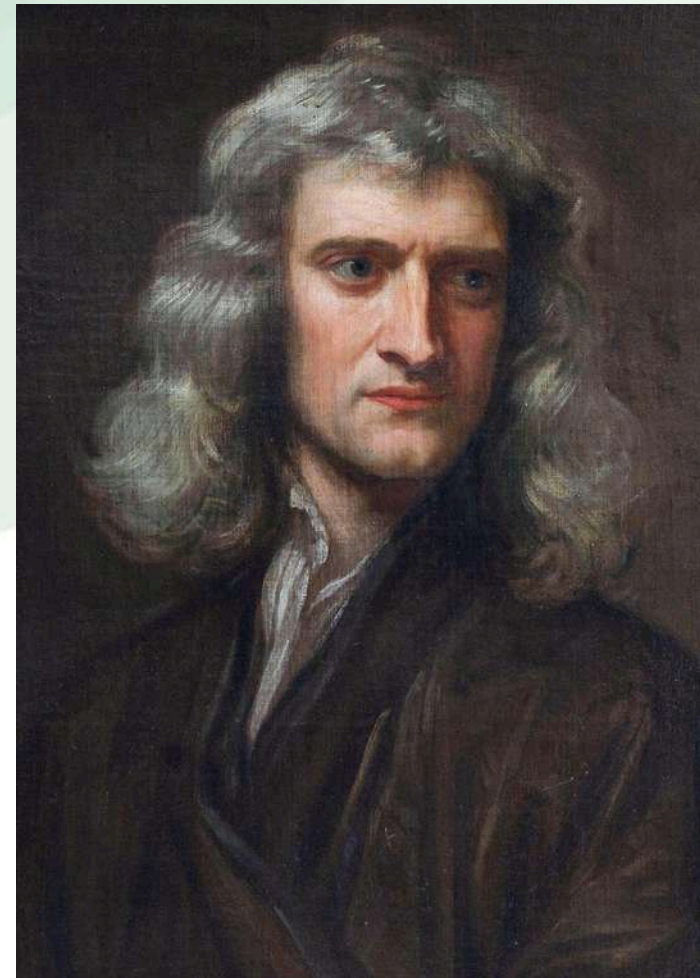
Source: www.nas.nasa.gov/SC17/demos/demo5.html



How Drones Generate Lift



$$L = \dot{m} \cdot \Delta v_y = \dot{m} \cdot w_{\text{downwash}}$$



Isaac Newton

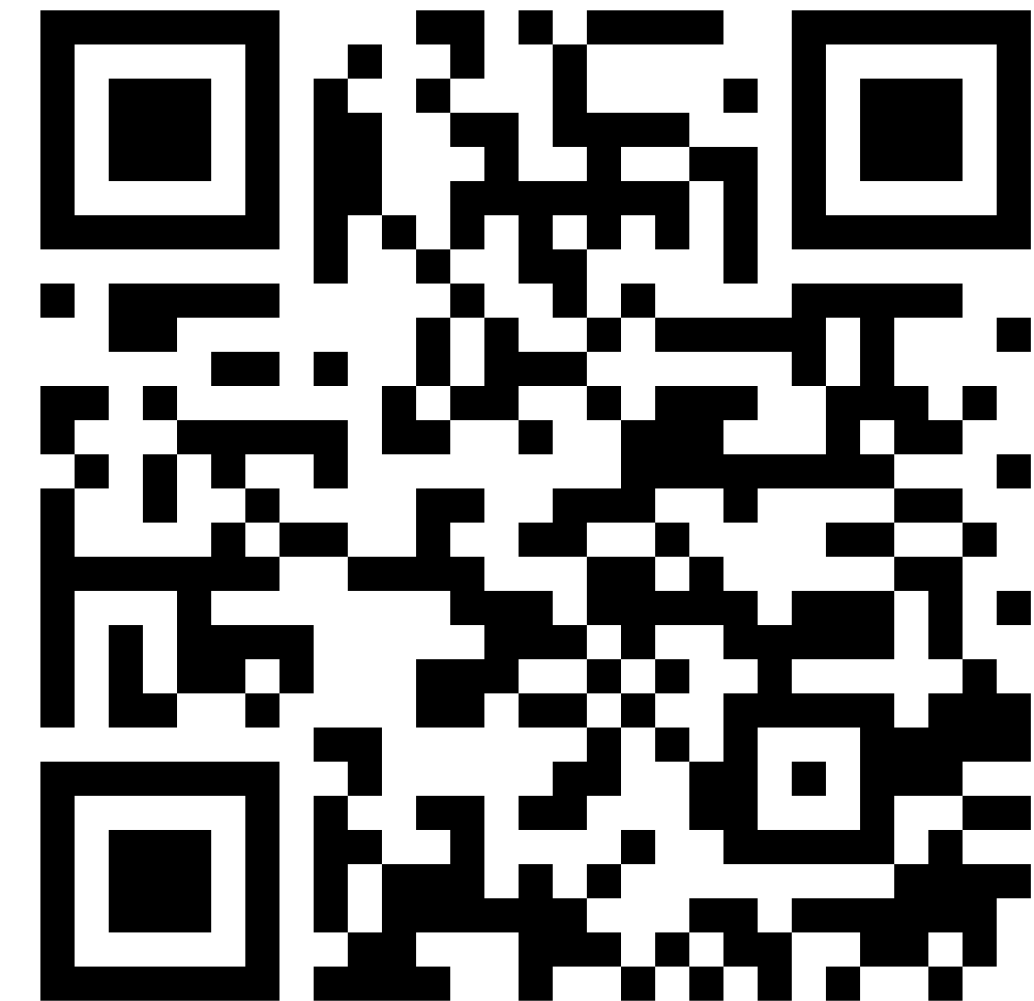
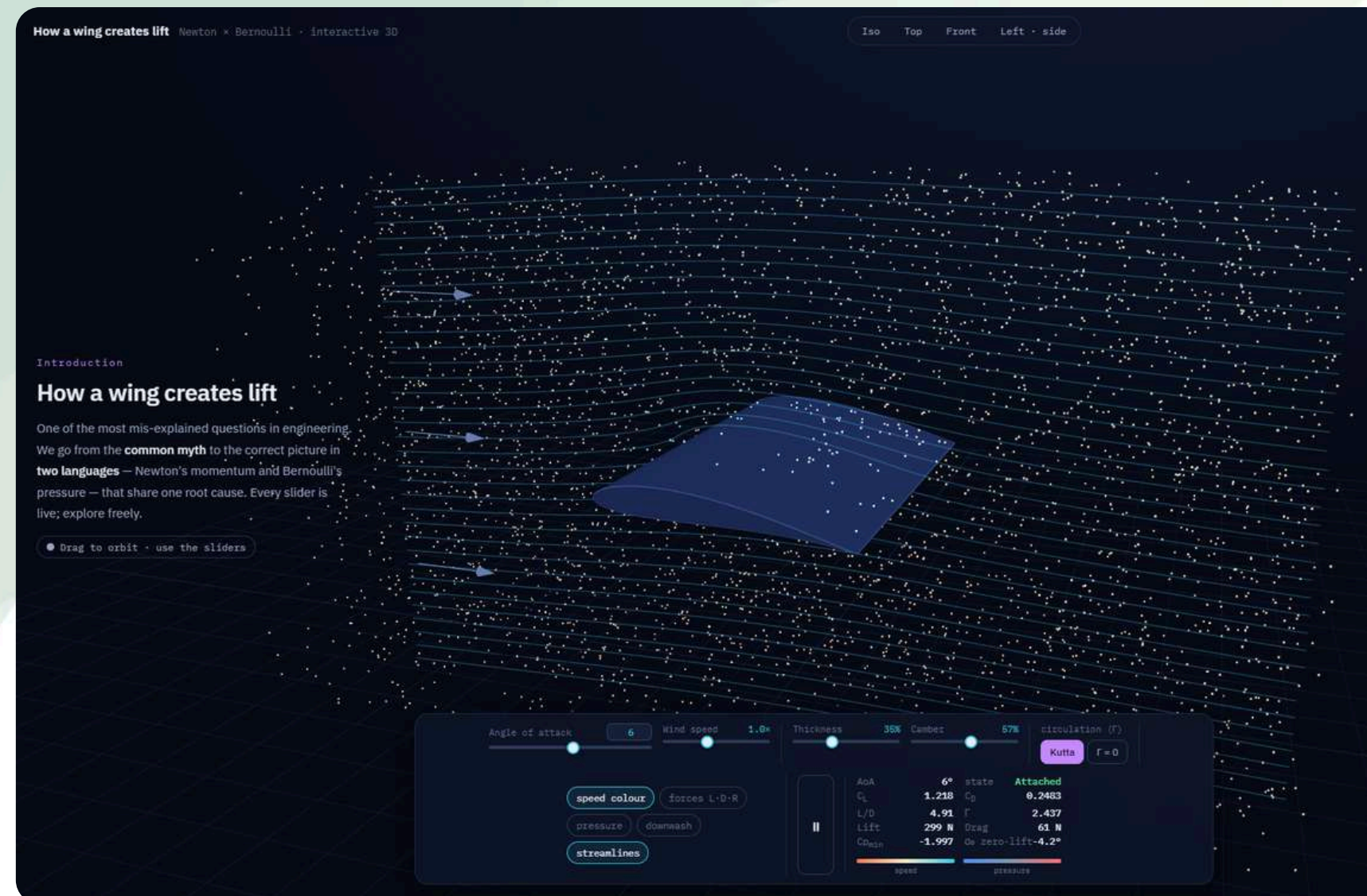


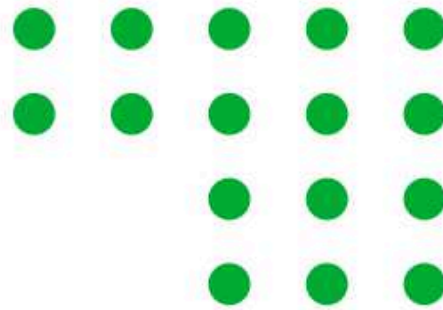
Daniel Bernoulli

$$p + \frac{1}{2}\rho v^2 = p_{\infty} + \frac{1}{2}\rho V_{\infty}^2 = \text{constant}$$



How Drones Generate Lift

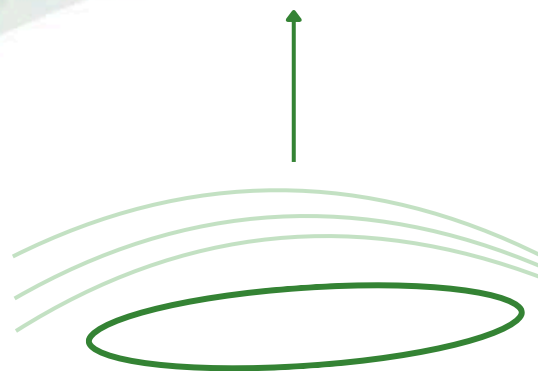




Force & Torque

Thrust

- Lift



Drag

- aerodynamic drag



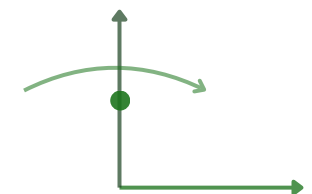
Gravity

- Weight

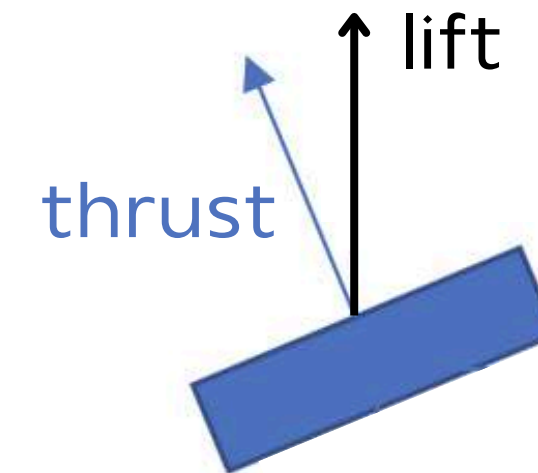
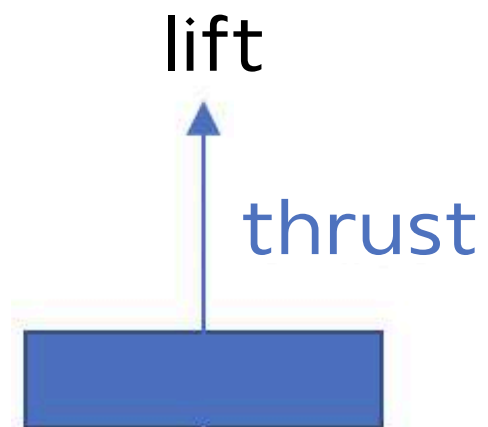
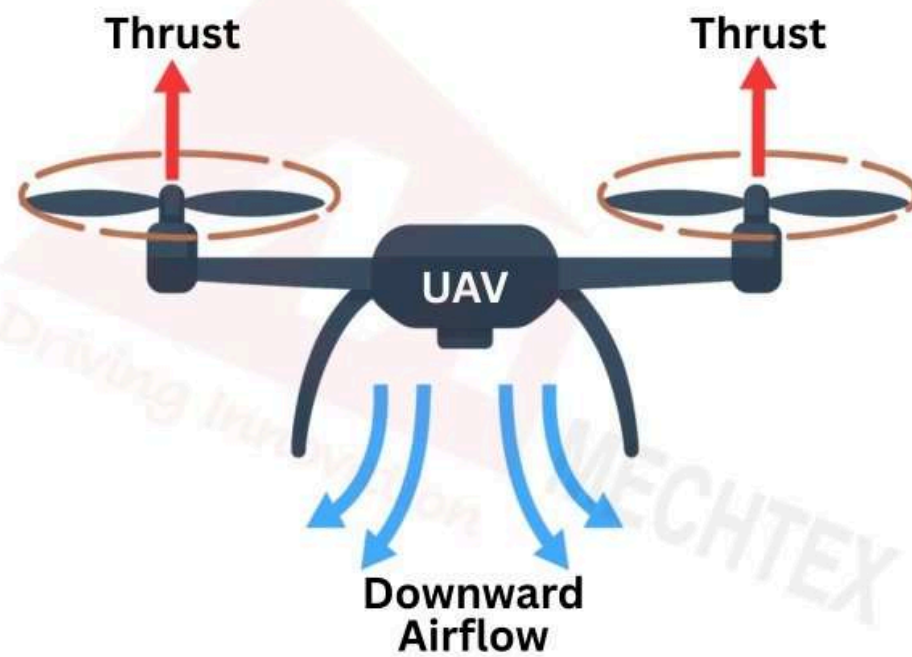


Torque

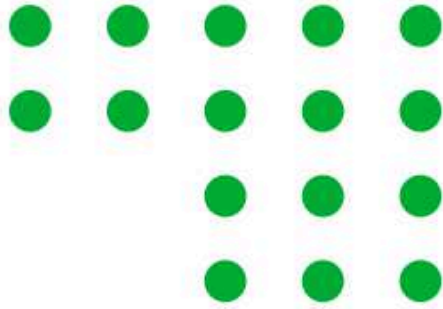
- Roll , pitch , yaw



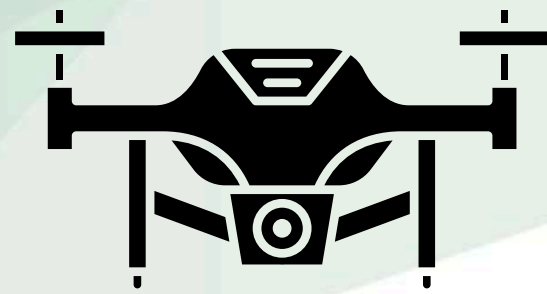
Lift Force



$$L = T \cdot \cos\phi \cdot \cos\theta$$

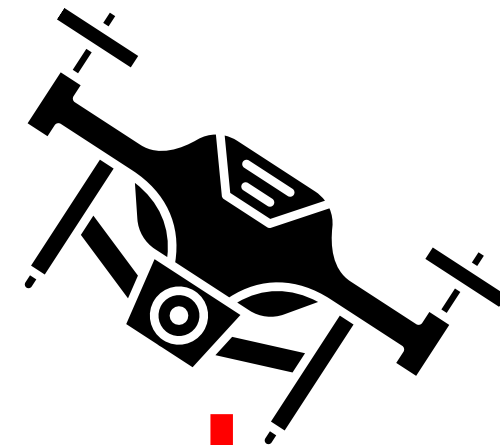


Weight



W

$$W = m \cdot g$$

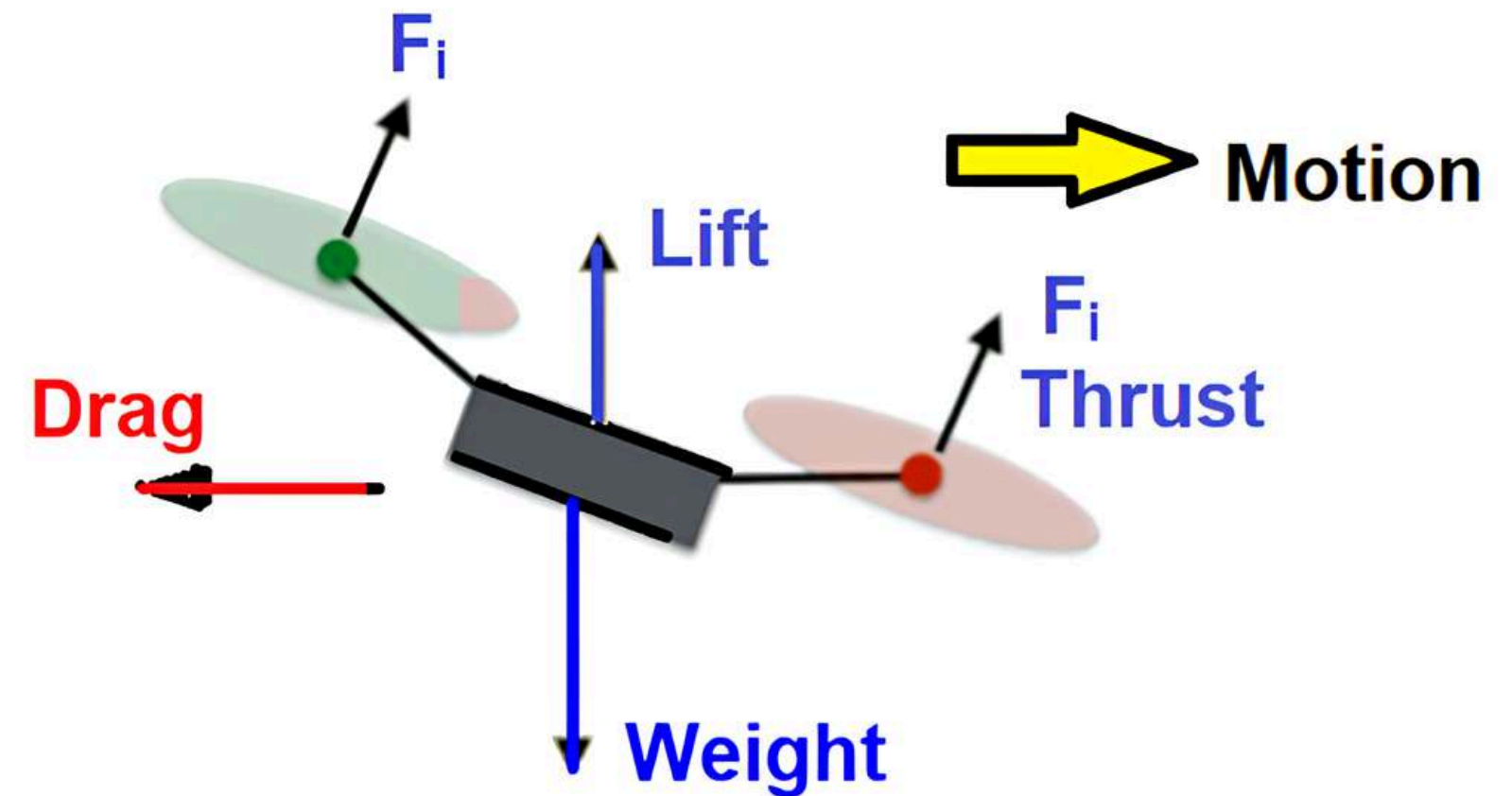


W



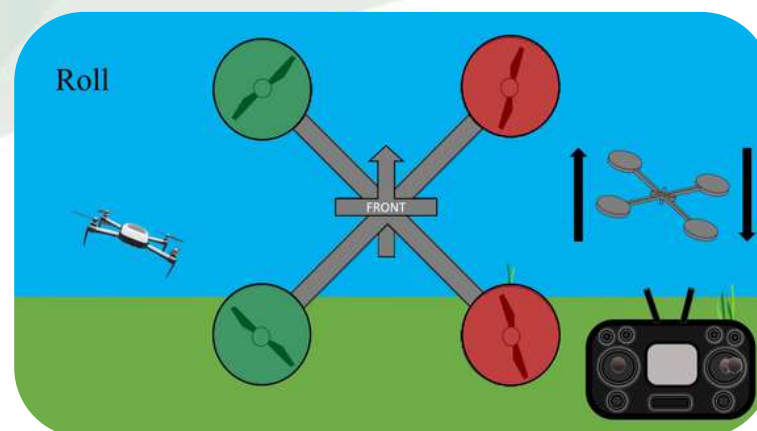
Drag force

$$F_D = \frac{1}{2} \rho v^2 C_D A$$

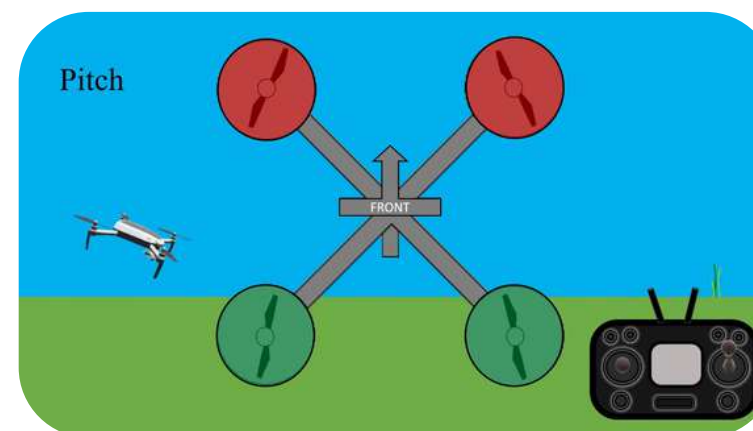


Torque

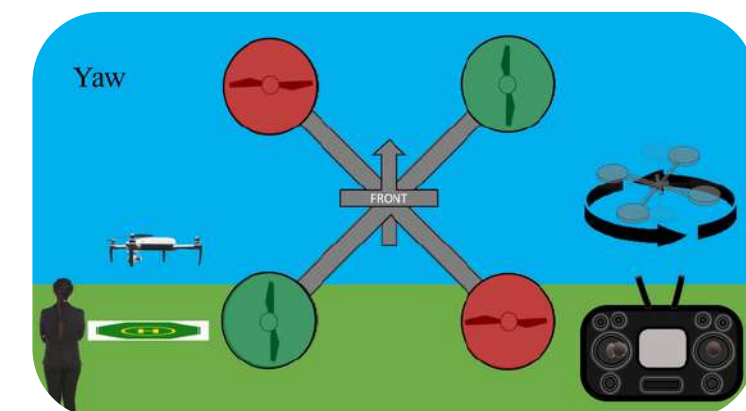
Roll Torque
(φ -X axis)



Pitch Torque
(θ -Y axis)

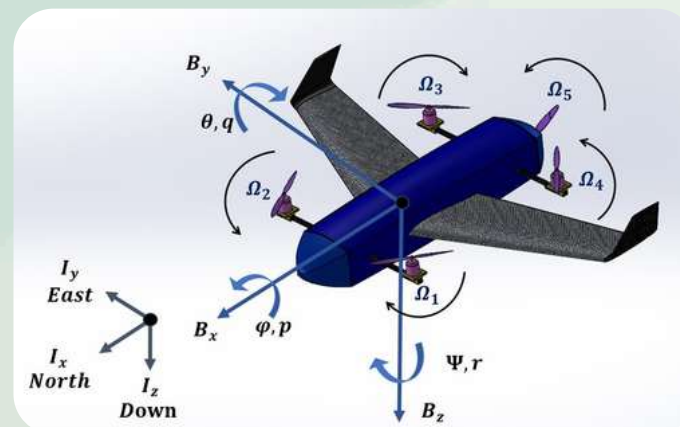


Yaw Torque
(ψ -Z axis)



Frames of Reference

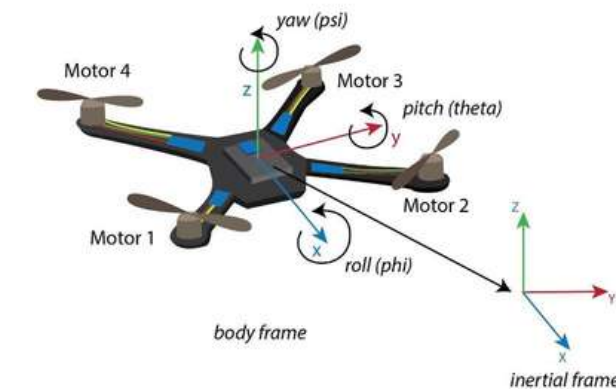
Inertial/Fixed Frame



Source: www.researchgate.net

Origin: launch location (on the Earth's surface).
Axes: fixed relative to Earth at that point in time (north-east-down, or some ENU/NED convention).
Use: good for describing trajectories in a "global" sense.

Body Frame



Source: www.researchgate.net

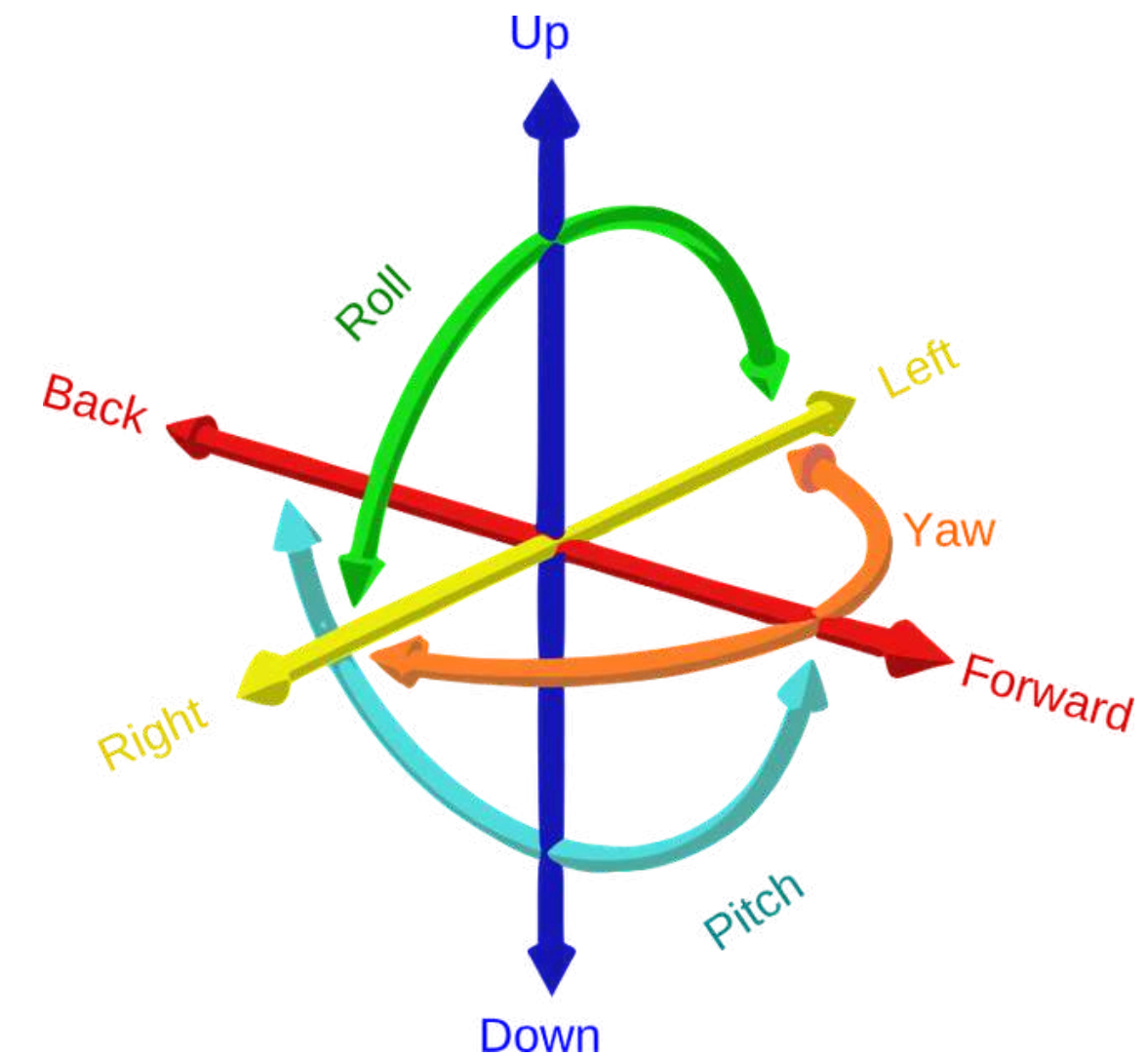
Origin: the vehicle's center of gravity (CoG).
Axes: aligned with the actual vehicle geometry — typically x forward, y right, z down (aircraft convention) or z up (spacecraft sometimes).
Use: sensors (IMU, gyros, accelerometers) and control forces naturally live here.



Degrees of Freedom (DOF)

6 DOF

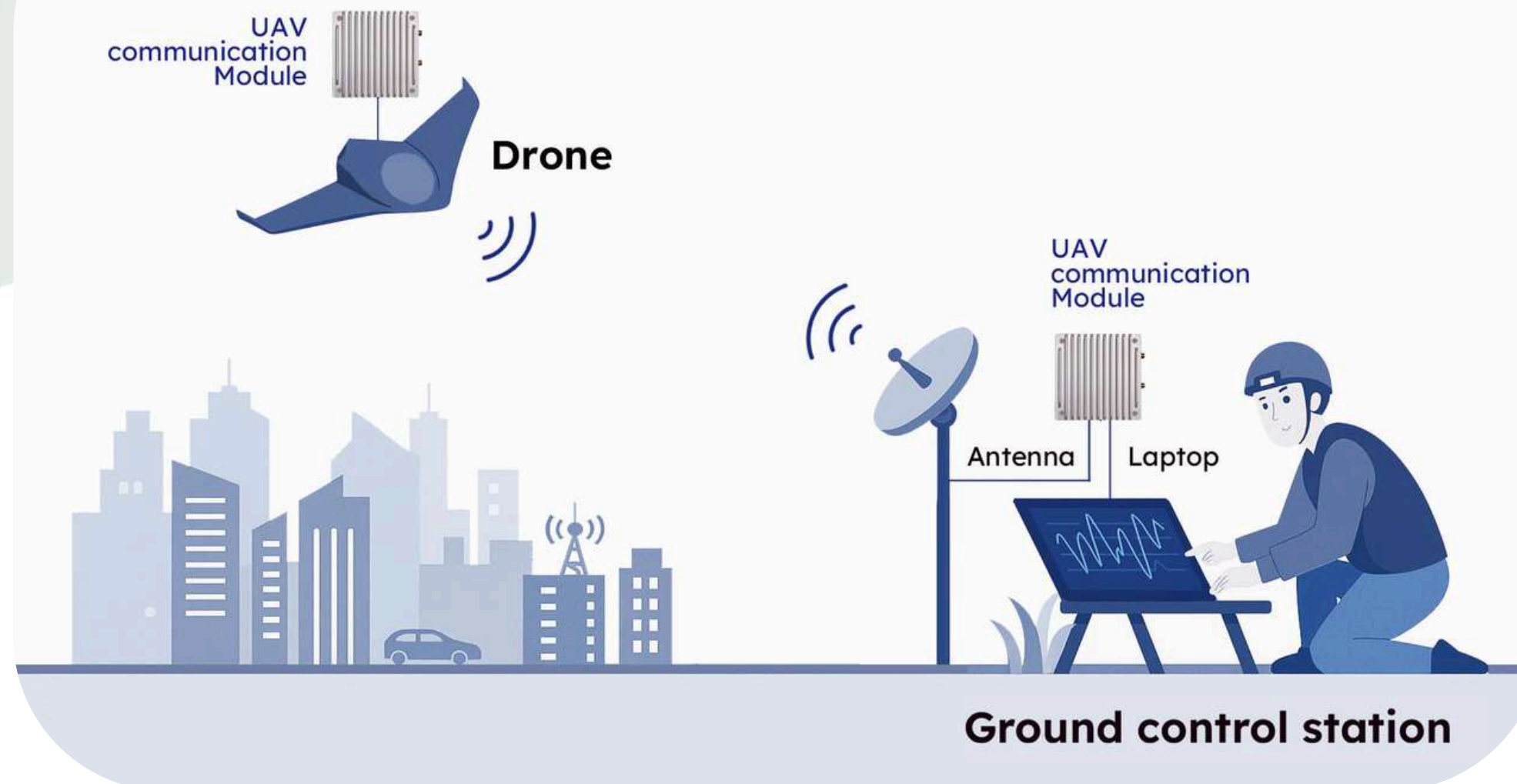
- Translational : x, y, z
- Rotational : roll (ϕ), pitch (θ), yaw (ψ)



Source: en.wikipedia.org

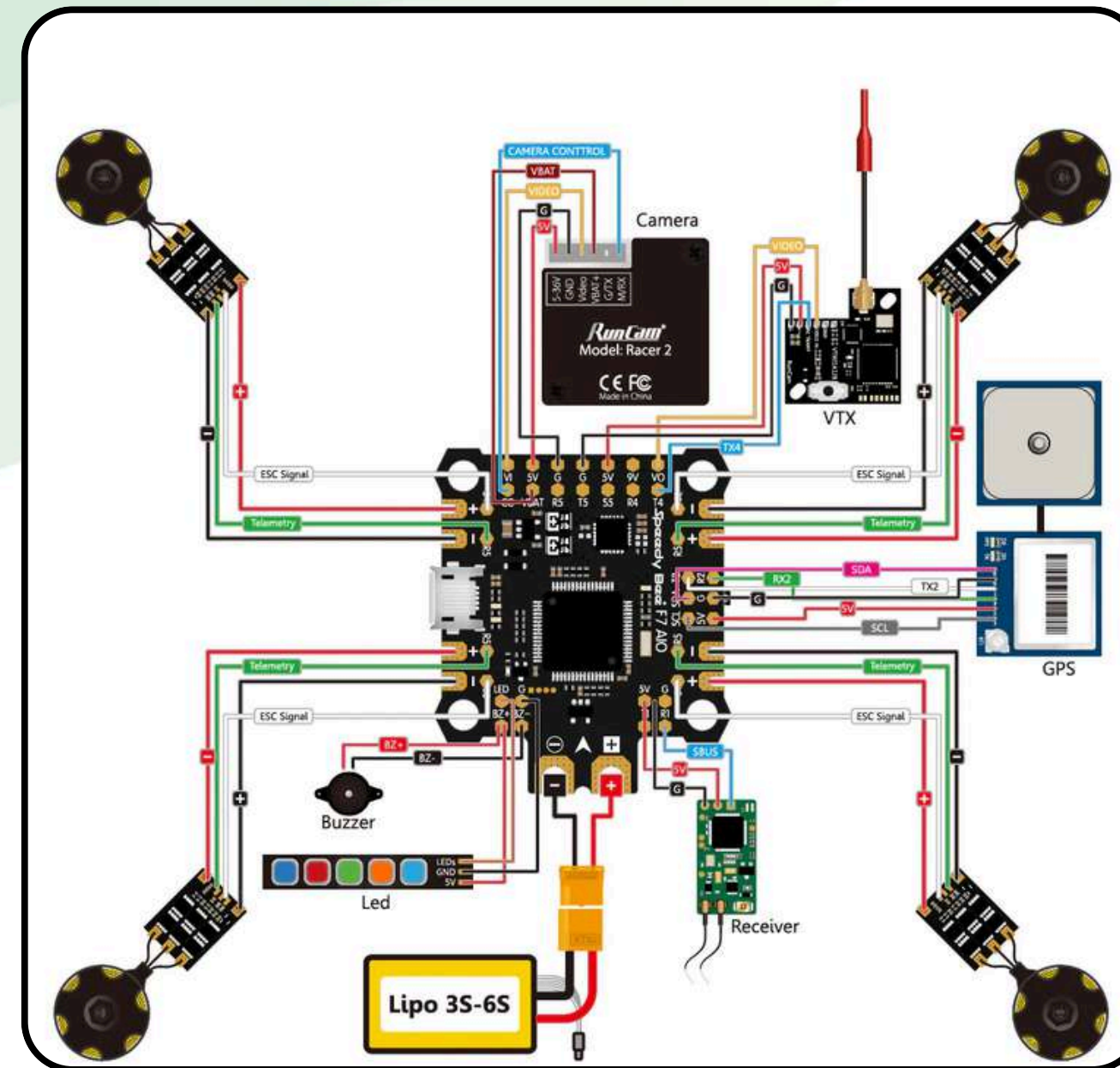


Communication System

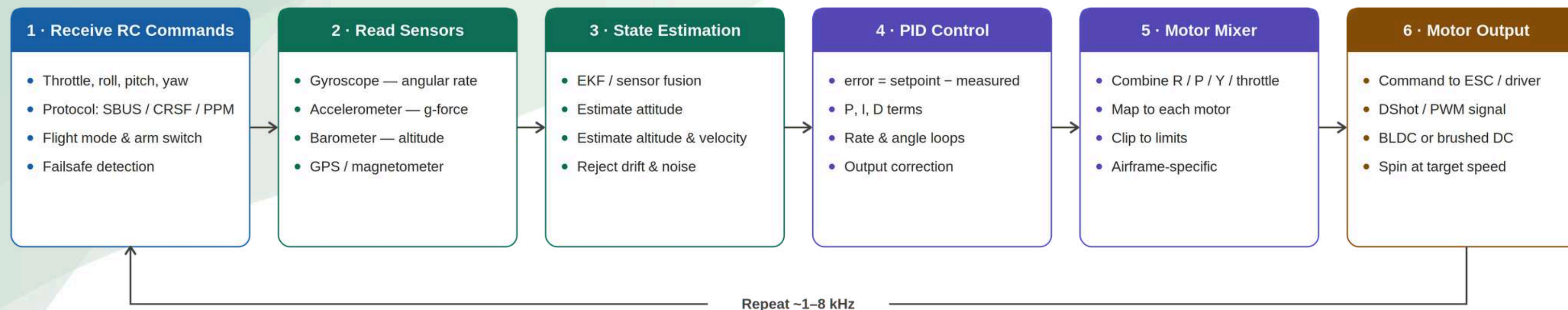
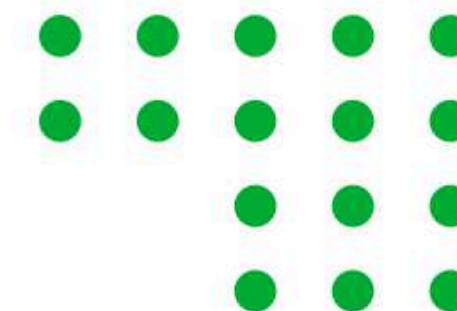


Source : <https://cfdflowengineering>

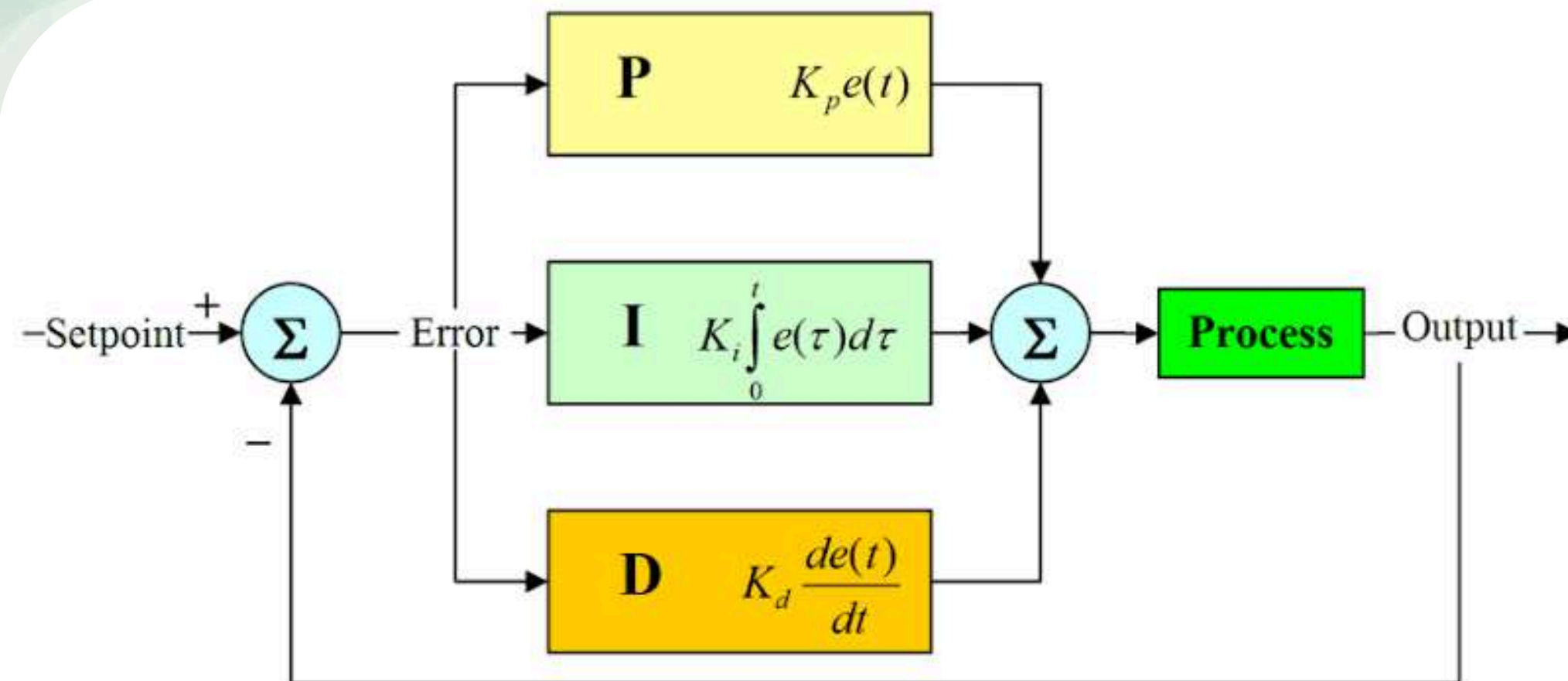
Flight Controller



Flight Control Loop



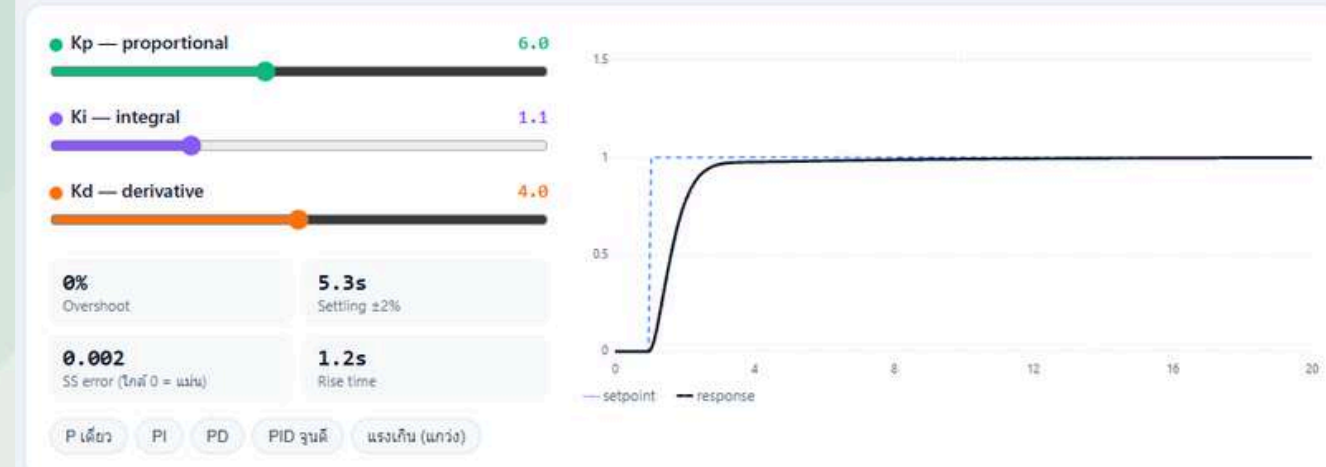
PID Control



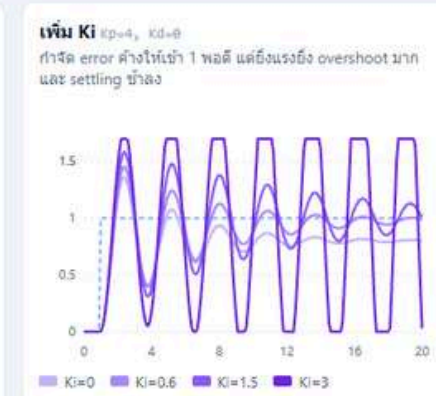
PID Control

PID Tuning Lab

Plant = mass-spring-damper · step response ที่ $t=1s$ · เปลี่ยน gain แล้วดูผลสด



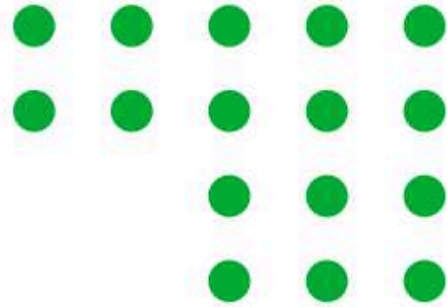
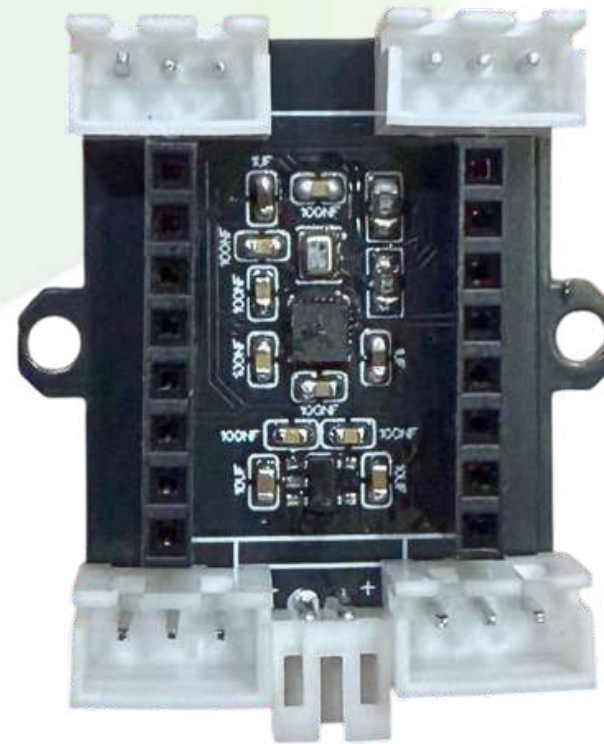
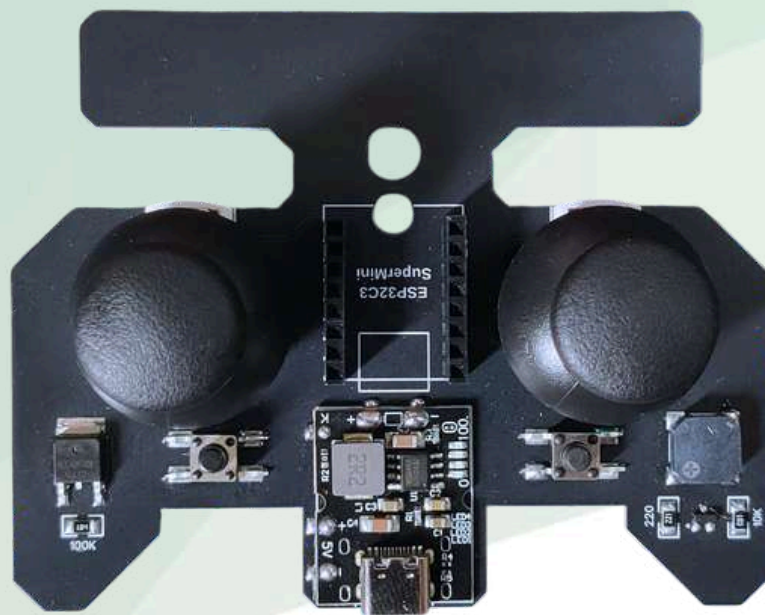
เพิ่มค่าแต่ละตัวขึ้นทีละขั้น — เกิดอะไรขึ้น

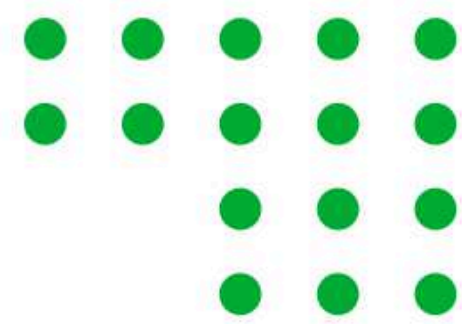


จำลองด้วย derivative-on-measurement (กิน derivative kick) เหมือน flight controller จริง - ไม่มี noise / anti-windup เพื่อให้เห็นผลของslac term ชัด



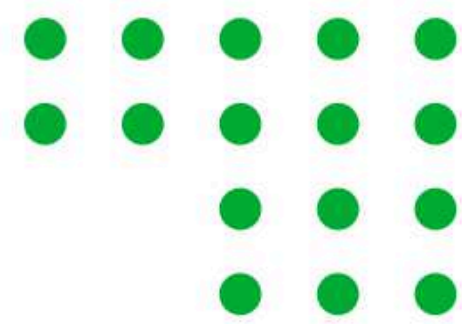
SimpleFly Learning Kit





Firmware Installation





Setup & Calibration

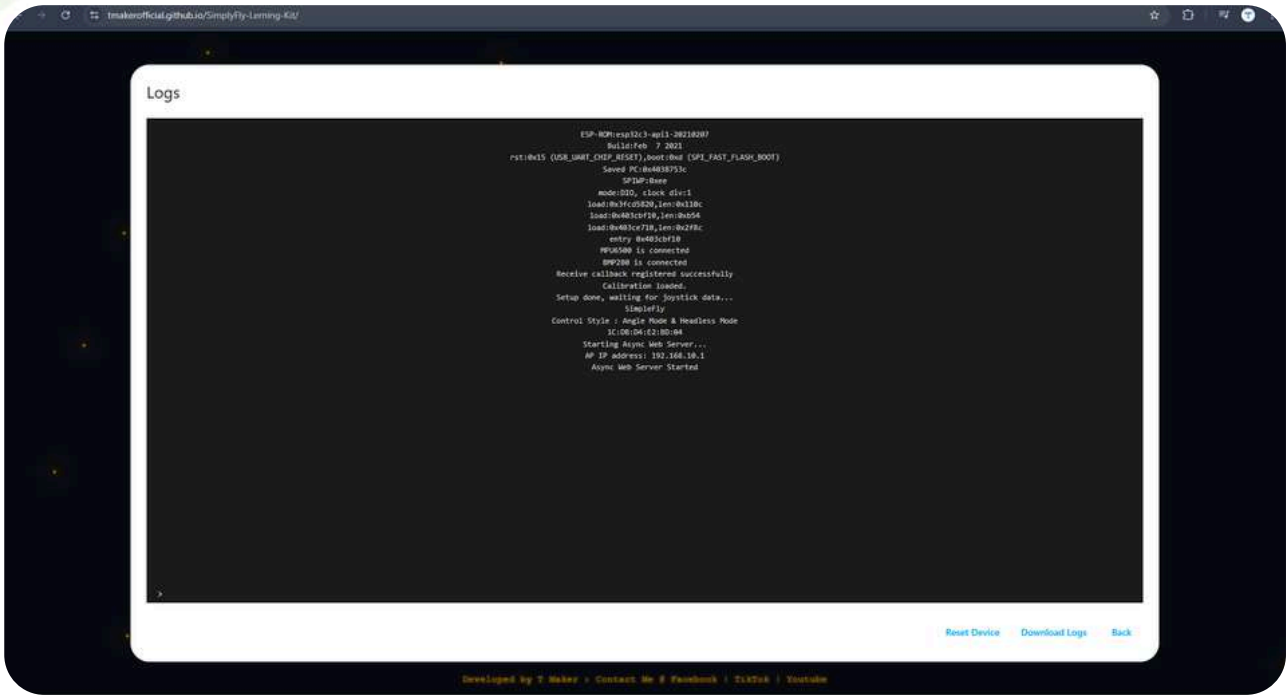


calibrate sensor



calibrate minmax
save minmax

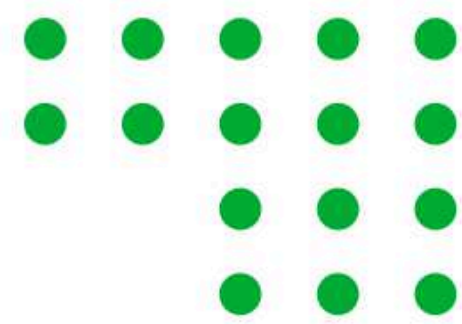
calibrate center
save center



set max = " MAC Address"



Tuning



PID Roll Rate

Kp

1.00

Ki

0.00

Kd

0.05

PID Pitch Rate

Kp

1.00

Ki

0.00

Kd

0.05

PID Yaw Rate

Kp

3.00

Ki

0.00

Kd

0.03

PID Altitude Rate (m/s)

Kp

70.00

Ki

0.00

Kd

3.50

PID Roll Angle

Kp

7.00

Ki

0.00

Kd

0.00

PID Pitch Angle

Kp

7.00

Ki

0.00

Kd

0.00

PID Yaw Angle

Kp

4.00

Ki

0.00

Kd

0.00

PID Altitude

Kp

1.00

Ki

0.00

Kd

0.00

Trim

Trim Roll

0.30

Trim Pitch

0.40

Trim Yaw

0.00

Trim Altitude

0.00

Base Speed

Base Speed

1400

UPDATE SETTINGS

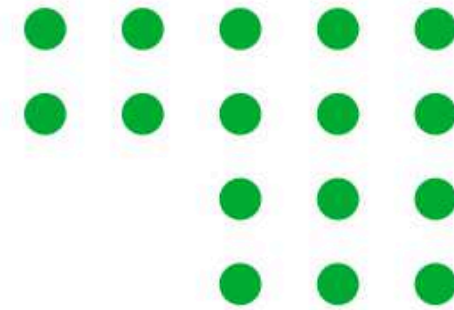
Calibration

CALIBRATE ACCEL & GYRO

RESET CALIBRATION



Tuning



min value of **XL** | Yaw rate

min value of **YL** | Altitude rate

min value of **XR** | Roll angle

min value of **YR** | Pitch angle

Save Min/Max
Calibrate Center
Reset All

Mapping Parameters

→	-20	20	←
→	-50	50	←
→	-15	15	←
→	15	-15	←

Save Mapping

Set Receiver MAC Address

Save MAC

max value of **XL** | Yaw rate

max value of **YL** | Altitude rate

max value of **XR** | Roll angle

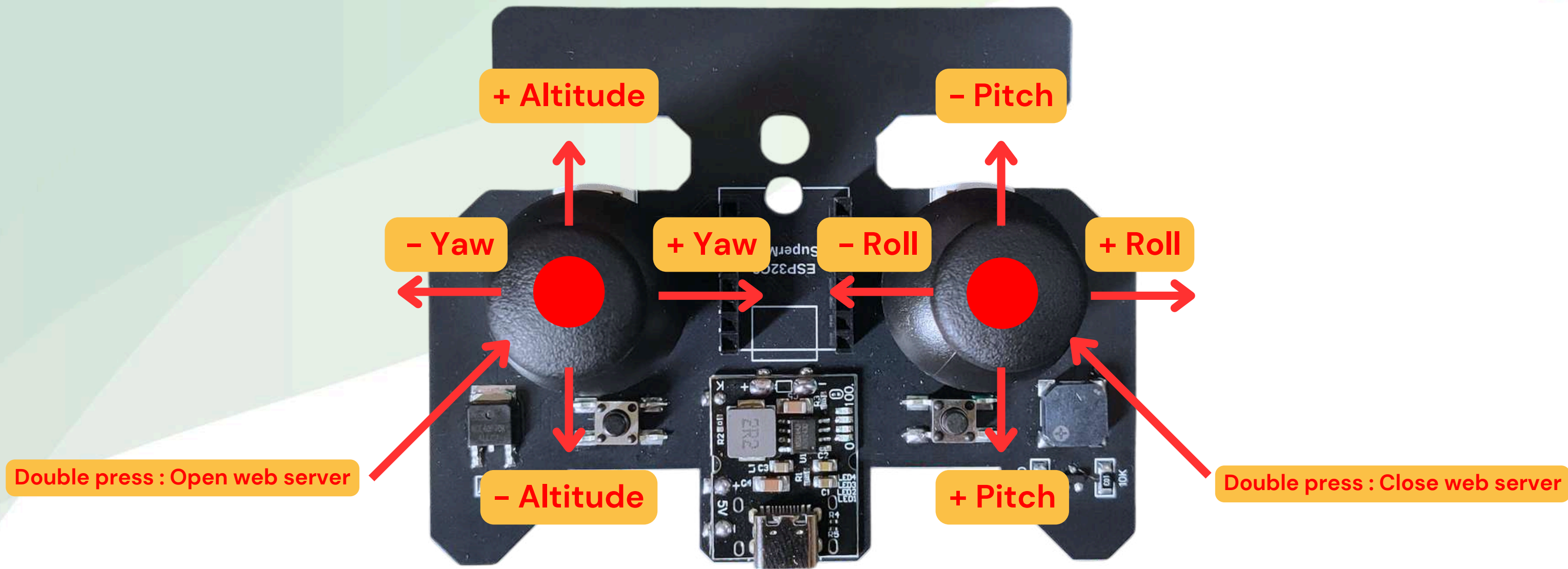
max value of **YR** | Pitch angle

Unit

- XL : deg/s
- YL : cm/s
- XR : deg
- YR : deg



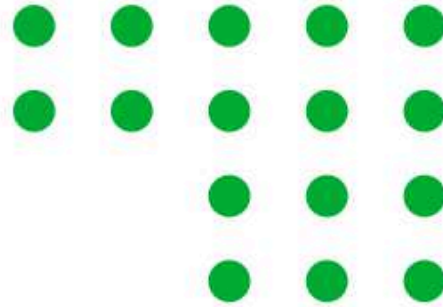
Manual Control



Back side : single press ==> power on | double press ==> power off

Press both the left and right joysticks simultaneously to toggle between Armed and Disarmed modes.





THANK YOU

T MAKER

